CRISTOBAL EYZAGUIRRE

Computer Science Ph.D. Student

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RESEARCH EXPERIENCE

Ph.D. Student

Stanford Vision and Learning Lab, Stanford University

September 2021 – Present | Stanford, CA

- Co-advised by Juan Carlos Niebles and Jiajun Wu on efficient video understanding, multimodal reasoning, and evaluation of generative video models.
- **ECCV 2026 (Under Review)**: Proposed *FastKV*, a dual-cache method for linear-time video prefill in long-video VLMs. **Stay tuned**.
- **ICML 2025**: Introduced a framework for estimating question complexity in VideoQA via visual program generation.
- **NeurIPS 2024**: Pioneered streaming video-language understanding by introducing an event-start detection benchmark, accompanying evaluation metrics, and dedicated modeling approaches.
- Co-authored papers on long-form video understanding, video grounding, and zero-shot optical flow extraction from generative video models (**CVPR 2025**, **NeurIPS 2024**, **NeurIPS 2025**).

Research Scientist Intern

Meta, GenAI Multimodal Team

Summer 2025 | Menlo Park, CA

- Worked on multimodal post-training for video understanding in Llama during a research internship in Meta's GenAI Multimodal team.

Research Intern

Toyota Research Institute

Summer 2023 | Los Gatos, CA

- Developed methods for characterizing question complexity in video question answering, later leading to **ICML 2025**.

Research Intern

Google Research

October 2020 – January 2021 | Mountain View, CA

- Researched active speaker detection from video and audio by integrating face detection, facial landmark extraction, and audio processing models into a MediaPipe pipeline.

Student Researcher

IALab, Pontificia Universidad Católica de Chile

July 2017 – June 2021 | Santiago, Chile

- Worked on adaptive computation for visual reasoning and language models, with publications at **CVPR 2020** and **ACL Workshop 2022**.

EDUCATION

Doctor of Philosophy (Ph.D.) in Computer Science

Stanford University

Sep. 2021 – Present | Stanford, CA

SVL Lab, co-advised by Juan Carlos Niebles and Jiajun Wu.

Emphasis on efficient Vision and Video Understanding.

M.Sc. Engineering

Pontificia Universidad Católica de Chile

July 2019 – June 2021 | Santiago, Chile

Supervised by Álvaro Soto, Ph.D. CMU.

Graduated with highest distinction.

B.Sc. Engineering

Pontificia Universidad Católica de Chile

Jan. 2015 – Dec. 2019 | Santiago, Chile

Major in Software Engineering, minor in Data Science. Graduated with distinction.

RESEARCH INTERESTS

Video understanding, multimodal learning, long-form video, generative video models, efficient inference.

TECHNICAL SKILLS

Programming

Python, C, C++, JavaScript.

ML / Research

PyTorch, NumPy, scikit-learn, TensorFlow, Hugging Face Transformers, lmms-eval.

Systems

Slurm, CUDA, Triton, vLLM, large-scale training, research codebases.

- Researched deep learning methods for visual reasoning, natural language processing, and video understanding under the supervision of Álvaro Soto.

TEACHING EXPERIENCE

- Stanford University: Course Assistant, CS231N: Deep Learning for Computer Vision (Spring 2026).
- Stanford University: Course Assistant, CS231N: Deep Learning for Computer Vision (Spring 2025).
- Pontificia Universidad Católica de Chile: Instructor, Professional Education Artificial Intelligence Program (2020–2021).
- Pontificia Universidad Católica de Chile: Teaching Assistant, Deep Learning (2020).
- Pontificia Universidad Católica de Chile: Teaching Assistant, Artificial Intelligence (2019–2020).
- Pontificia Universidad Católica de Chile: Coach and Product Manager, ABET-certified Capstone Project (2019).

SELECTED PUBLICATIONS

- **Linear Scaling Video VLMs for Long Video Understanding** (*FastKV*): Introduced a dual-cache inference method for linear-time video prefill in long-video VLMs. **ECCV 2026 Under Review**. **Stay tuned**.
- **Understanding Complexity in VideoQA via Visual Program Generation**. **ICML 2025**.
- **Streaming Detection of Queried Event Start**. **NeurIPS 2024** Datasets and Benchmarks Track.
- **DACT-BERT: Differentiable Adaptive Computation Time for an Efficient BERT Inference**. **ACL Workshop 2022**.
- **Differentiable Adaptive Computation Time for Visual Reasoning**. **CVPR 2020**.
- **Taming Generative Video Models for Zero-Shot Optical Flow Extraction**. **NeurIPS 2025**.
- **Revisiting the "Video" in Video-Language Understanding**. **CVPR 2022 Oral**.